

# Operating Systems for Intelligent Systems (ENGS2306) - Practical Lab Outline

## Course Information

- **Course Name:** Operating Systems for Intelligent Systems
- **Course Code:** ENGS2306
- **Semester:** Second Semester 2024/2025
- **Lecturer:** Eng. Mohammed E. Alhams
- **Duration:** 14 weeks
- **Delivery Method:** Pre-recorded lectures via YouTube and Moodle
- **Evaluation:** Assignments (60%), Final Exam (40%)

## Course Description

This course explores the role of operating systems in managing computers and intelligent devices, covering memory management (virtual and physical), interrupt routines, process and thread management, multiprocessing, I/O management, and an introduction to distributed, networked, and mobile operating systems (e.g., Android). The practical component focuses on hands-on exploration of Linux internals, studying and practicing Bash scripting, a theoretical comparison of Android, Unix, and Linux, and using VMware for virtualization.

## Practical Component Objectives

- Understand Linux OS internals and resource management (processes, memory, I/O, kernel, file system, users).
- Study and practice Bash scripting for system automation and management.
- Compare Android, Unix, and Linux theoretically, focusing on their relationship and design.
- Use VMware to explore OS environments, including Raspberry Pi OS.

## Weekly Plan

| Week | Topic                                         | Assignments/Tasks                                                                                                  |
|------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1    | Introduction to Linux in VMware               | Install Ubuntu 20.04 LTS in VMware; explore commands (ls, cd, man, etc).                                           |
| 2    | Linux File System and Permissions             | Assignment 1: Analyze file structures; use chmod, chown for permissions.                                           |
| 3    | Process Management in Linux                   | Assignment 2: Monitor processes with ps, top, htop.                                                                |
| 4    | Memory Management Exploration                 | Use free, vmstat to analyze memory usage; document findings.                                                       |
| 5    | Introduction to Bash Scripting                | Assignment 3: Practice basic Bash scripts for file management (e.g., find, grep); explain functions.               |
| 6    | Advanced Bash Scripting                       | Assignment 4: Practice Bash scripts for process monitoring/automation (e.g., loops, conditionals); document logic. |
| 7    | Interrupt Handling Analysis                   | Use dmesg to study kernel logs; document interrupt-related entries.                                                |
| 8    | I/O Management in Linux                       | Explore iostat, lsof; document I/O operations in VMware.                                                           |
| 9    | Theoretical Study: Android vs. Unix vs. Linux | Assignment 5: Compare Android, Unix, Linux (architecture, kernel, features).                                       |
| 10   | Multiprocessing and Threads                   | Analyze top output for multi-core usage; document thread behavior.                                                 |

|    |                              |                                                                                                      |
|----|------------------------------|------------------------------------------------------------------------------------------------------|
| 11 | Distributed Systems Overview | Set up multiple Ubuntu VMs in VMware; analyze network interactions with ping, netstat.               |
| 12 | Raspberry Pi OS Case Study   | Create a Raspberry Pi OS VM in VMware; document features (e.g., file system, I/O).                   |
| 13 | Review and Mini-Project      | Mini-project: Design a Bash script to analyze system logs (e.g., /var/log/syslog) or monitor system. |
| 14 | Final Practical Exam         |                                                                                                      |

## Required Tools and Installation Instructions

Lectures are provided via YouTube and Moodle, and essential resources must be downloaded from the internet. Ensure a stable connection when accessing these materials. Below are the required tools and installation instructions:

- **Ubuntu 20.04 LTS:**
  - Download ISO from Ubuntu official website or Moodle link.
  - Install in VMware Workstation Player.
  - Minimum requirements: 4GB RAM, 20GB storage.
  - Installation: Boot ISO in VMware, follow Ubuntu installer, select "Erase disk and install Ubuntu."
- **Raspberry Pi OS:**
  - Download ISO (Raspberry Pi OS Lite, 64-bit) from Raspberry Pi official website or Moodle link.
  - Install in VMware Workstation Player.
  - Minimum requirements: 2GB RAM, 10GB storage.
  - Installation: Create a new VM in VMware, select "Other Linux 64-bit" as the OS type, attach the Raspberry Pi OS ISO, and follow the installer.

## Assessment Details

- **Assignments (60%):** Five assignments (12% each) covering Linux commands, Bash script practice, process/memory monitoring, I/O analysis, and OS comparisons.
- **Final Practical Exam (40%):** Assessment details to be provided by the lecturer.

## Notes for Students

- Access pre-recorded lectures via YouTube and Moodle; ensure a stable internet connection for streaming or downloading.
- Download all required software (Ubuntu ISO, Raspberry Pi OS ISO) from the provided Moodle links or official websites.
- Ensure a computer with at least 4GB RAM and 20GB free storage for Ubuntu, and 2GB RAM and 10GB for Raspberry Pi OS.
- Contact Eng. Mohammed E. Alhams for technical support via Moodle or the course coordinator.